

Marine Pollution in the United Arab Emirates: Current Status, Future Challenges, and Mitigation Strategies

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Abstract

The marine environment of the United Arab Emirates (UAE) is a vital national asset, supporting biodiversity, desalination, and a thriving tourism industry. However, this ecosystem faces increasing pressure from various forms of pollution, including microplastics, industrial runoff, and the impacts of desalination. This article provides a comprehensive analysis of marine pollution in the UAE, with a focus on emerging challenges and future mitigation strategies for the 2024-2026 period. It examines the prevalence of microplastics in Abu Dhabi's waters, the ecological consequences of brine discharge from desalination plants, and the impact of coastal development on sensitive habitats like mangroves and coral reefs. The research also highlights the UAE's proactive regulatory response, including the expansion of Marine Protected Areas (MPAs) and the integration of advanced monitoring technologies such as AI and satellite imaging. By synthesizing recent scientific data and policy initiatives, the paper outlines a roadmap for the sustainable management of the UAE's marine resources. The findings emphasize the need for a holistic approach that combines stringent industrial regulations with community-based conservation efforts to ensure the long-term health of the Arabian Gulf and the Sea of Oman.

Keywords: Marine Pollution, UAE, Microplastics, Desalination, Brine Discharge, Coastal Development, Marine Protected Areas, Sustainability, Arabian Gulf.

1. Introduction

The marine waters of the United Arab Emirates (UAE), encompassing the Arabian Gulf and the Sea of Oman, are among the most unique and stressed marine environments in the world. Characterized by high salinity, extreme temperatures, and limited water exchange, these ecosystems are particularly vulnerable to anthropogenic pressures. As the UAE continues its rapid economic and urban expansion, the protection of its marine environment has become a critical national priority (u.ae, 2025).

Marine pollution in the UAE manifests in several forms, from visible plastic debris to invisible chemical contaminants and thermal pollution. The consequences of these pollutants are far-reaching, affecting not only marine biodiversity but also the quality of water used for desalination—the nation’s primary source of fresh water. In recent years, the emergence of microplastics as a significant threat has added a new layer of complexity to marine conservation efforts (Frontiers in Marine Science, 2026).

This article explores the current state of marine pollution in the UAE, identifying the primary sources of contamination and their ecological impacts. It also examines the future challenges that the nation must address to safeguard its marine heritage. Furthermore, the paper discusses the innovative strategies and policy frameworks being implemented to mitigate pollution and promote the sustainable use of marine resources. Through this analysis, the article aims to provide a detailed overview of the UAE’s commitment to marine environmental health in the 21st century.

2. Emerging Threats: Microplastics and Chemical Contaminants

One of the most pressing challenges facing the UAE’s marine environment is the proliferation of microplastics—plastic particles smaller than 5mm. These pollutants are persistent, difficult to remove, and can enter the marine food web, posing risks to both wildlife and human health.

2.1. Microplastic Prevalence in UAE Waters

Recent studies have highlighted a high prevalence of microplastics in the marine waters and sediments of the Abu Dhabi Emirate. Research conducted in 2025-2026

identified microplastics ranging from 300 to 5,000 μm , with significant concentrations found near industrial zones and urban centers (Frontiers in Marine Science, 2026). These particles originate from various sources, including the breakdown of larger plastic waste, industrial processes, and synthetic fibers from wastewater. The unique hydrodynamics of the Arabian Gulf can lead to the accumulation of these particles in specific coastal areas, further intensifying their impact on local ecosystems.

2.2. Industrial and Urban Runoff

The UAE's coastal areas are hubs of industrial activity and urban development. Runoff from these areas can carry a variety of contaminants into the sea, including heavy metals, hydrocarbons, and nutrients from fertilizers. While the UAE has implemented strict regulations on industrial discharge, the cumulative effect of small-scale runoff from urban infrastructure remains a challenge. These pollutants can lead to eutrophication—a process where excess nutrients cause algal blooms that deplete oxygen in the water, leading to “dead zones” where marine life cannot survive.

3. The Impact of Desalination and Coastal Development

The UAE's reliance on the sea for water and land has significant environmental trade-offs. Desalination and coastal engineering are two of the most impactful activities on the marine environment.

3.1. Brine Discharge and Thermal Pollution

Desalination is essential for the UAE's survival, but the process produces brine—a highly concentrated salt solution that is typically discharged back into the sea. This brine is often warmer than the surrounding seawater and contains chemicals used in the desalination process. The discharge of brine can increase local salinity and temperature, stressing marine organisms, particularly coral reefs and seagrass beds (Yousef & Mac Dowell, 2026). Future strategies are focusing on “Zero Liquid Discharge” (ZLD) technologies and the use of solar energy to power desalination, which can mitigate both the chemical and carbon footprint of the process.

3.2. Coastal Reclamation and Habitat Loss

To accommodate its growing population and tourism industry, the UAE has engaged in extensive coastal reclamation and the construction of artificial islands. While these projects are engineering marvels, they can lead to the direct loss of sensitive habitats such as mangroves, seagrass meadows, and coral reefs. These habitats are crucial for marine biodiversity, acting as nurseries for fish and providing coastal protection. The UAE is now prioritizing “nature-based solutions,” such as the restoration of mangroves and the creation of artificial reefs, to offset the impacts of coastal development (u.ae, 2025).

4. Mitigation Strategies and Policy Frameworks

The UAE government has adopted a proactive and multi-layered approach to marine conservation, combining regulation, technology, and international cooperation.

4.1. Expansion of Marine Protected Areas (MPAs)

The UAE has significantly expanded its network of Marine Protected Areas (MPAs) to safeguard critical habitats and species. These areas, such as the Marawah Marine Biosphere Reserve in Abu Dhabi and the Jebel Ali Marine Sanctuary in Dubai, are subject to strict regulations on fishing, shipping, and development. By 2026, the UAE aims to increase the percentage of its marine territory under protection, aligning with global biodiversity targets (u.ae, 2025).

4.2. Advanced Monitoring and AI Integration

Technology is playing a vital role in marine environmental management. The UAE is utilizing satellite imaging and AI-powered sensors to monitor water quality, track oil spills, and detect illegal fishing activities in real-time. These tools allow for a more rapid and effective response to pollution incidents. Furthermore, AI is being used to model the dispersion of brine and other pollutants, helping planners to minimize the environmental impact of new coastal projects (Albreem et al., 2023).

4.3. The 2026 Plastics Policy and Marine Health

The comprehensive 2026 Plastics Policy, which includes a ban on various single-use plastics and mandates for circularity, is expected to have a direct and positive impact on marine health. By reducing the amount of plastic waste generated on land, the UAE is cutting off a major source of marine debris and microplastics (Entrepreneur MENA, 2026).

5. Future Challenges and the Path Forward

Despite the robust policy framework, several challenges remain for the future of marine conservation in the UAE.

- **Climate Change and Ocean Warming:** Rising sea temperatures and ocean acidification pose a long-term threat to the Arabian Gulf's ecosystems, particularly its coral reefs. Mitigation efforts must be coupled with global action on climate change.
- **Regional Cooperation:** Marine pollution is a transboundary issue. Effective management of the Arabian Gulf requires close cooperation with neighboring countries to address shared challenges such as oil spills and maritime traffic.
- **Sustainable Desalination:** As water demand grows, finding ways to desalinate water without harming the marine environment is a critical technological and environmental challenge.

The path forward for the UAE involves a continued commitment to science-based management and innovation. By integrating marine conservation into national development plans and fostering a culture of environmental responsibility, the UAE can ensure that its marine waters remain a source of life and prosperity for generations to come.

6. Conclusion

The marine environment of the United Arab Emirates is a precious but fragile resource that faces significant pressure from pollution and development. From the emerging threat of microplastics to the long-standing impacts of desalination and coastal

reclamation, the challenges are complex and multifaceted. However, the UAE's proactive response—characterized by the expansion of protected areas, the adoption of advanced monitoring technologies, and the implementation of ambitious circular economy policies—provides a strong foundation for marine conservation. The next few years will be critical as the nation scales up its efforts to mitigate pollution and restore degraded habitats. By continuing to lead in environmental innovation and regional cooperation, the UAE can safeguard its marine heritage and demonstrate that sustainable development and marine health are inextricably linked. The future of the UAE's seas depends on the collective efforts of the government, the private sector, and the community to value and protect this vital blue heart of the nation.